

Electrical Fault Detection

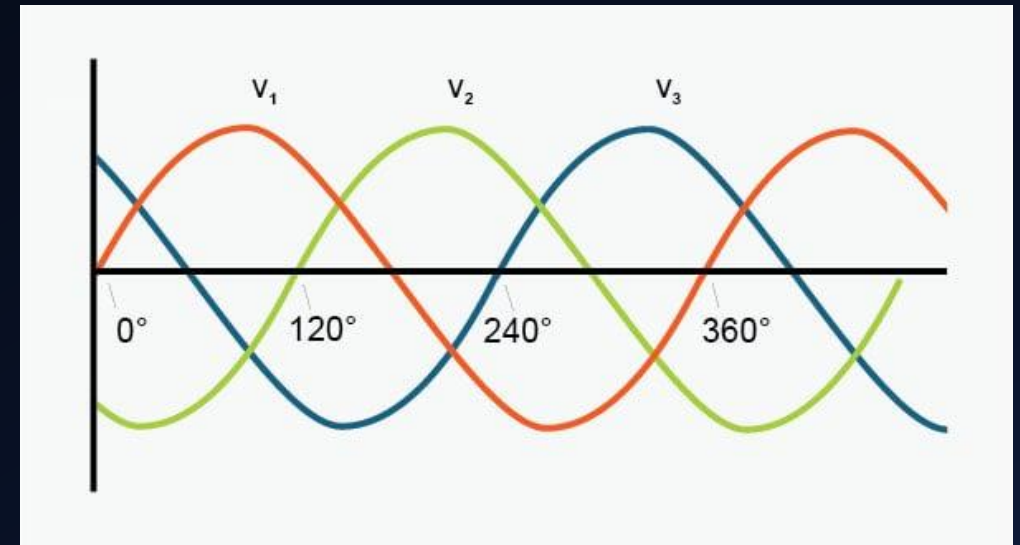
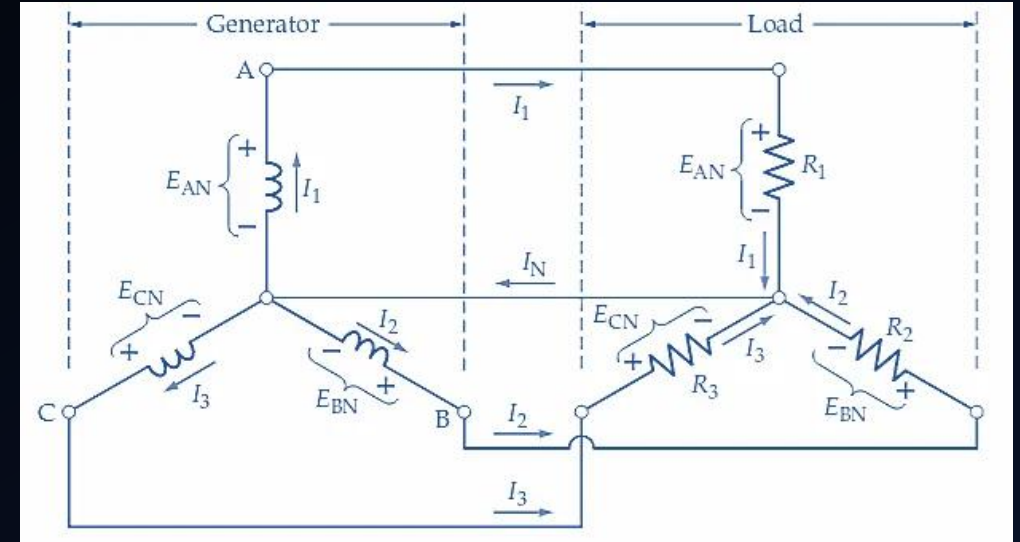
CS4320 CAPSTONE – LEVI DOCKSTADER

PROBLEM STATEMENT

- DEFINITION: USE MACHINE LEARNING TO DETECT AND CLASSIFY ELECTRICAL FAULTS IN A SIMULATED 3-PHASE GRID SYSTEM USING ONLY VOLTAGE AND CURRENT VALUES
- TASK TYPE: MULTICLASS CLASSIFICATION, THOUGH ORIGINALLY BINARY WAS USED
- APPLICATION: SUPPORT DECISIONS ABOUT GRID MANAGEMENT BY HELPING PREDICT AND DETECT ISSUES USING NON-TRANSIENT DATA

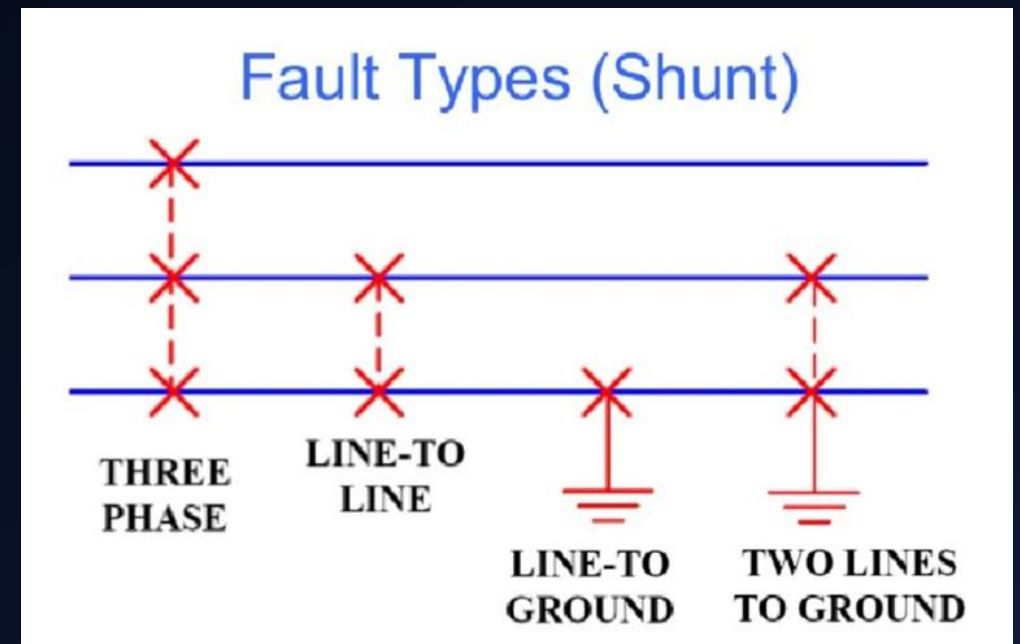
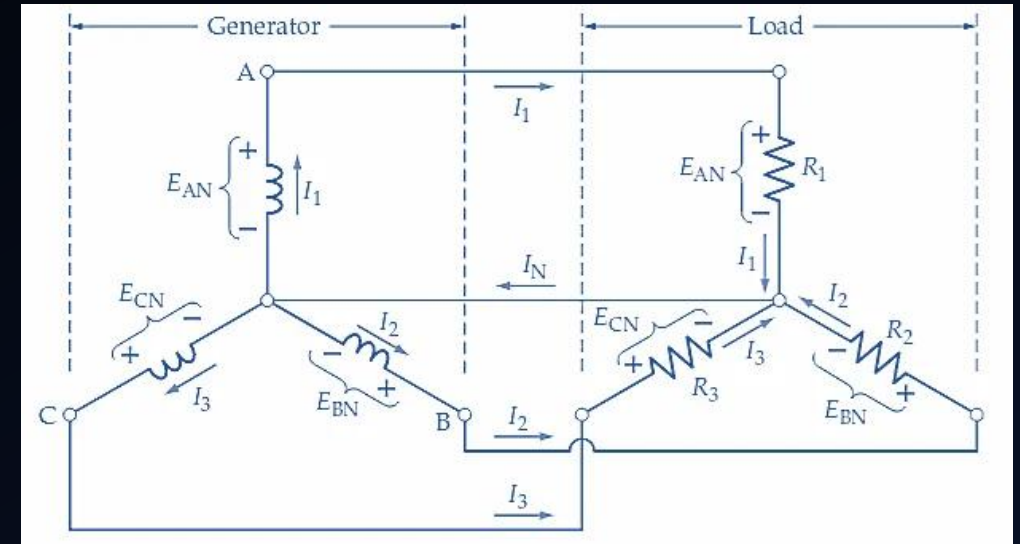
Background: 3 Phase Circuits

- 3 phase – 3 sine waves (ABC) offset from the others by 120°
- Ground – Common reference shared by all phases
- Ohm's Law: $V = IZ$
 - V: Voltage (V)
 - I: Current (A)
 - Z: Impedance (Ω)



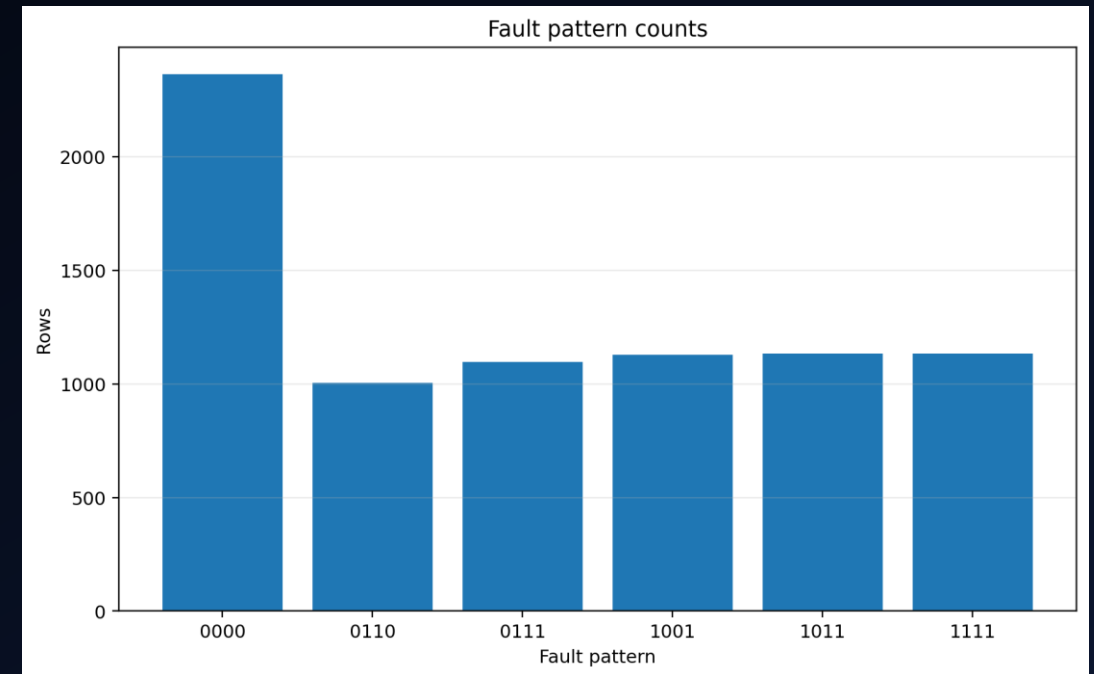
Background: Grid Faults

- During a fault, a line is somehow damaged, causing it to either touch another line or the ground.
- Faults cause electrical values to change from what's expected.
 - Typically, voltage and impedance decrease while current increases.

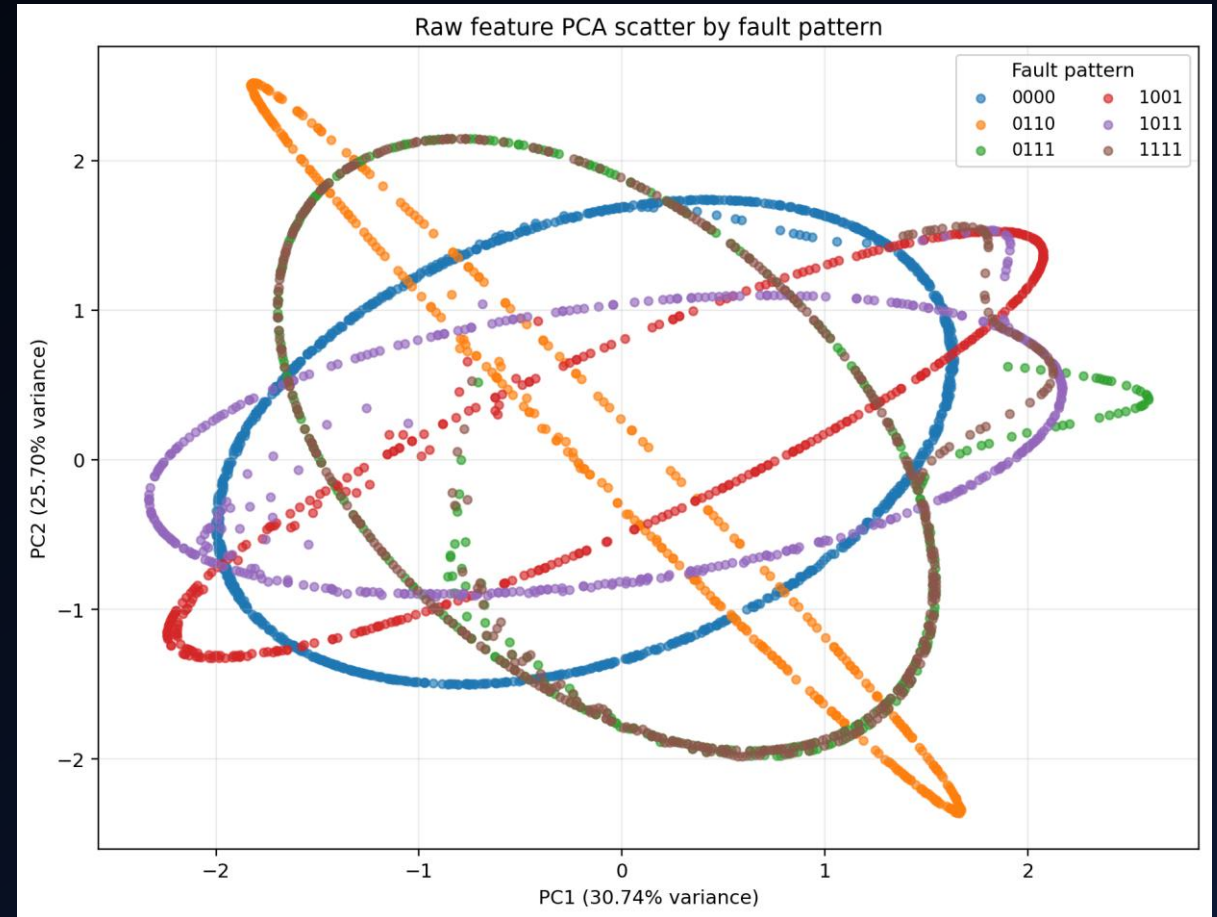
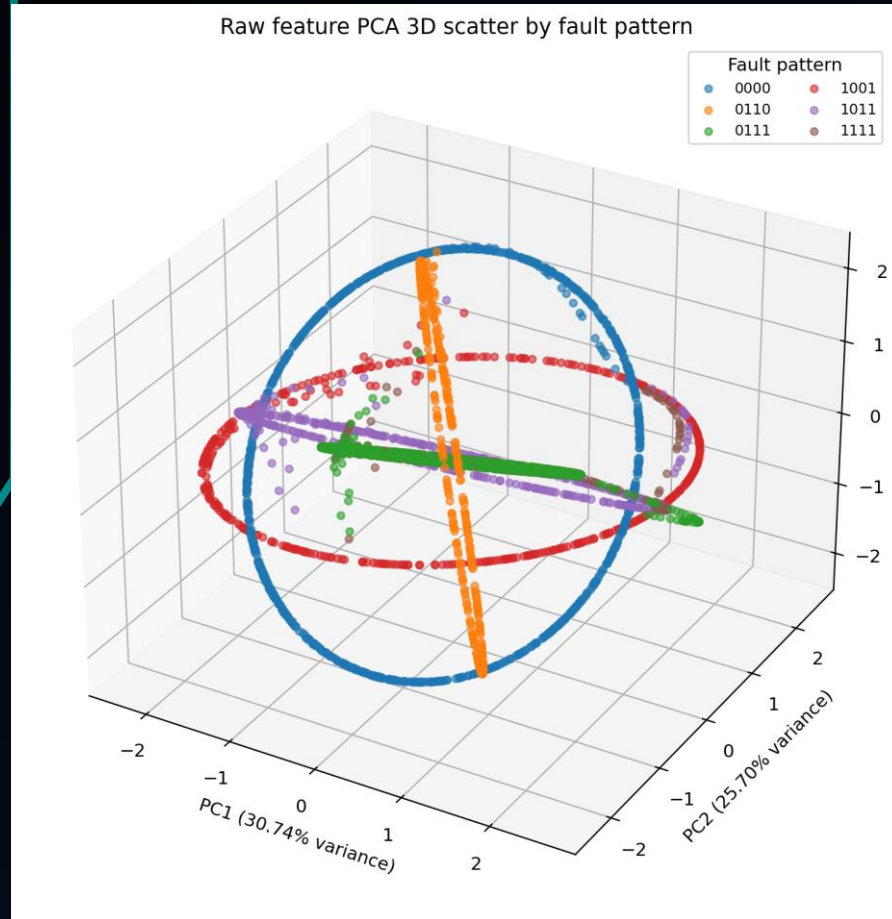


- Input:
 - 6 raw features
 - 3 voltage values
 - 3 current values
 - Labels excluded to prevent leakage
 - 7681 rows
- Output:
 - 4-digit-binary fault classifier prediction
 - GCBA 0000-1111
 - Early experiments performed binary fault classification
- Feature engineering:
 - 10 proxy features
 - Each proxy is related to a data point's impedance and phase.
 - Proxies were only developed during my final model experimenting.

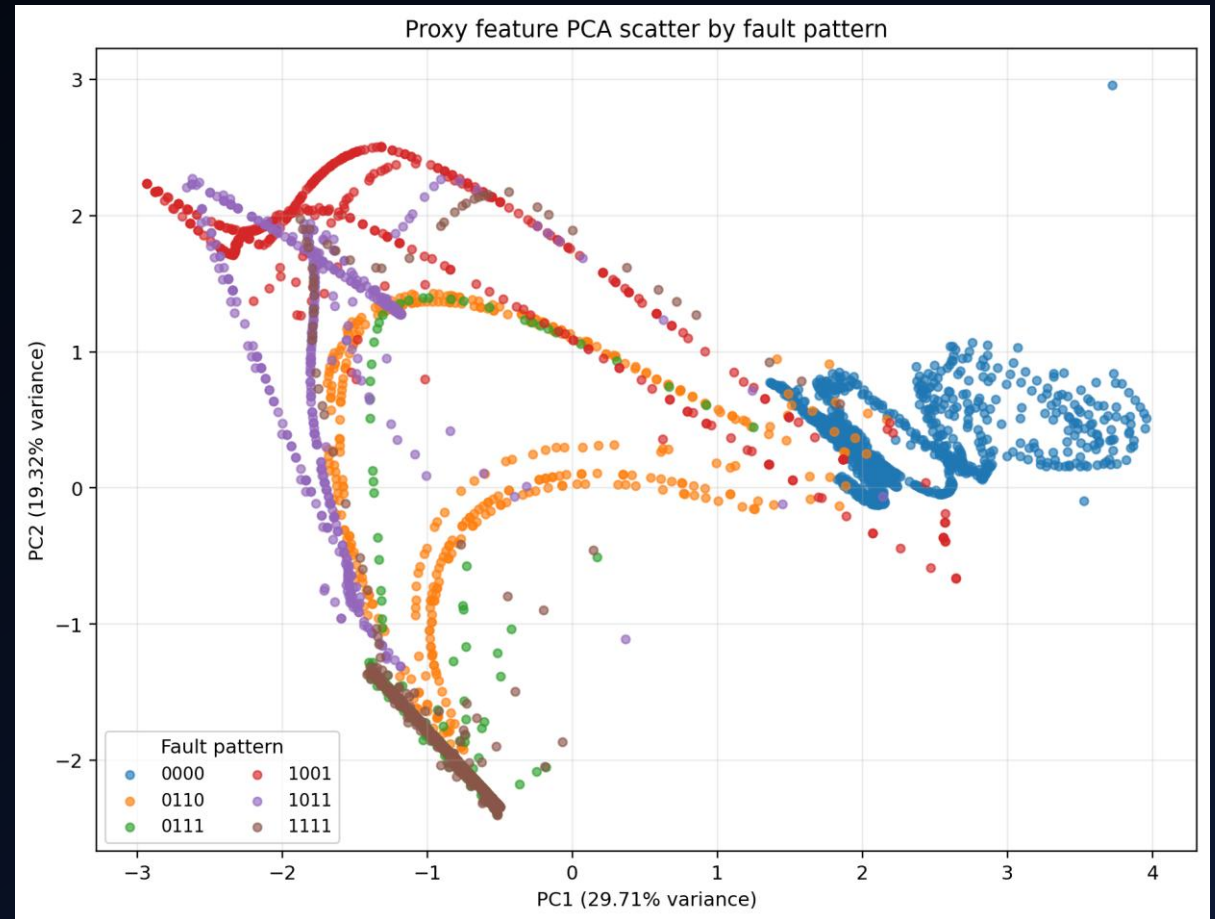
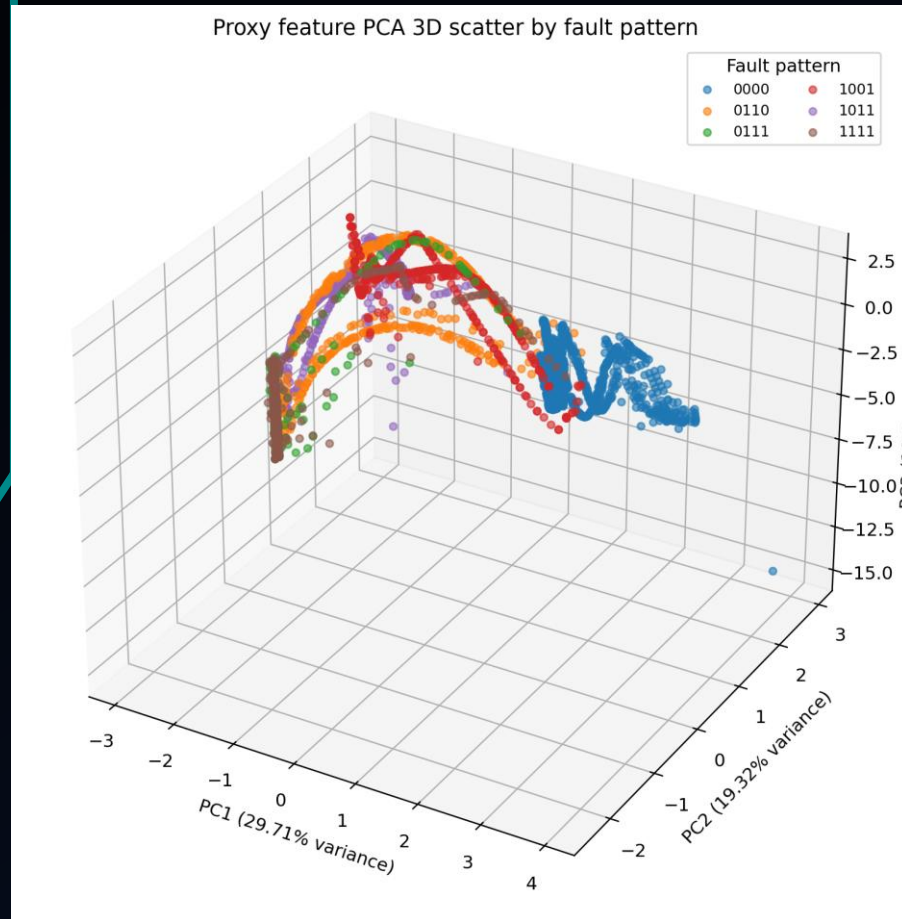
APPROACH AND ARCHITECTURE



Searching for Structure – Raw Feature Visualization



Searching for Structure – Proxy Feature Visualization



MODEL SELECTION

Key considerations

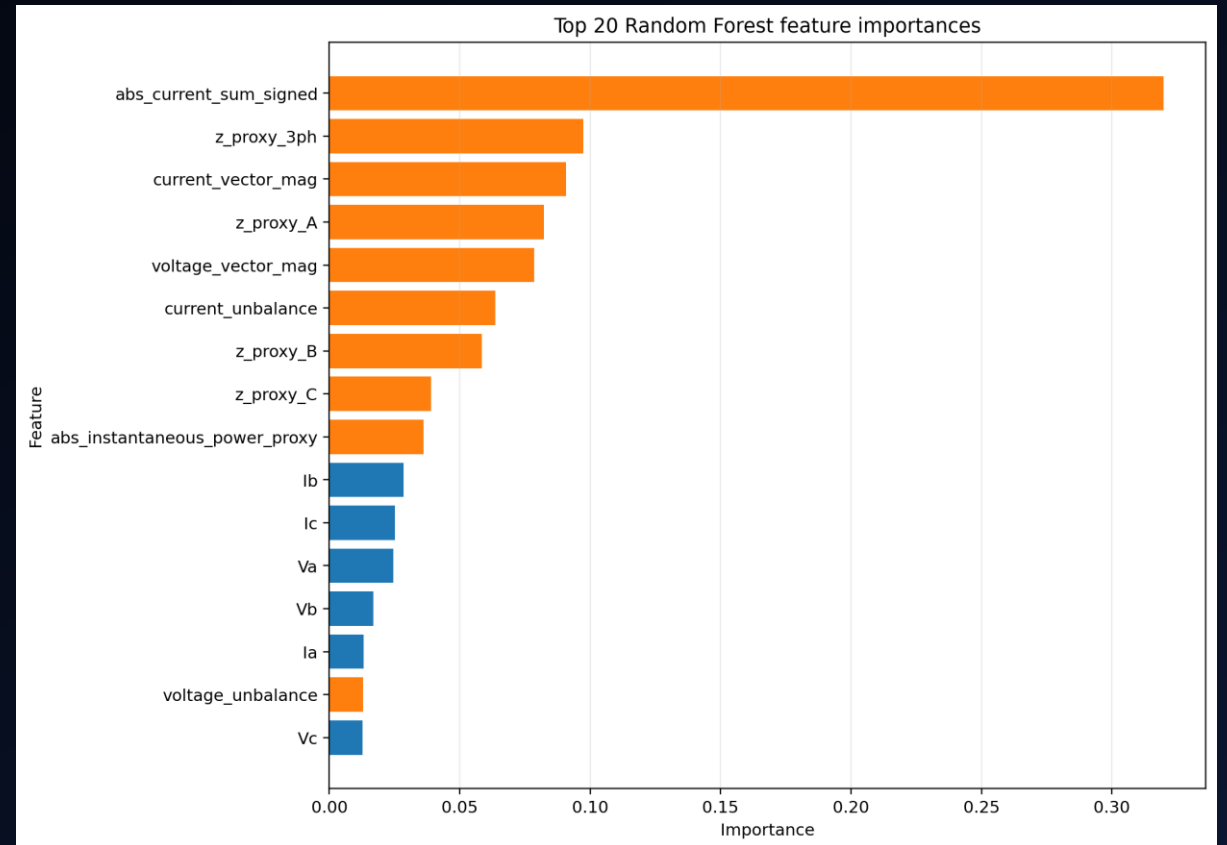
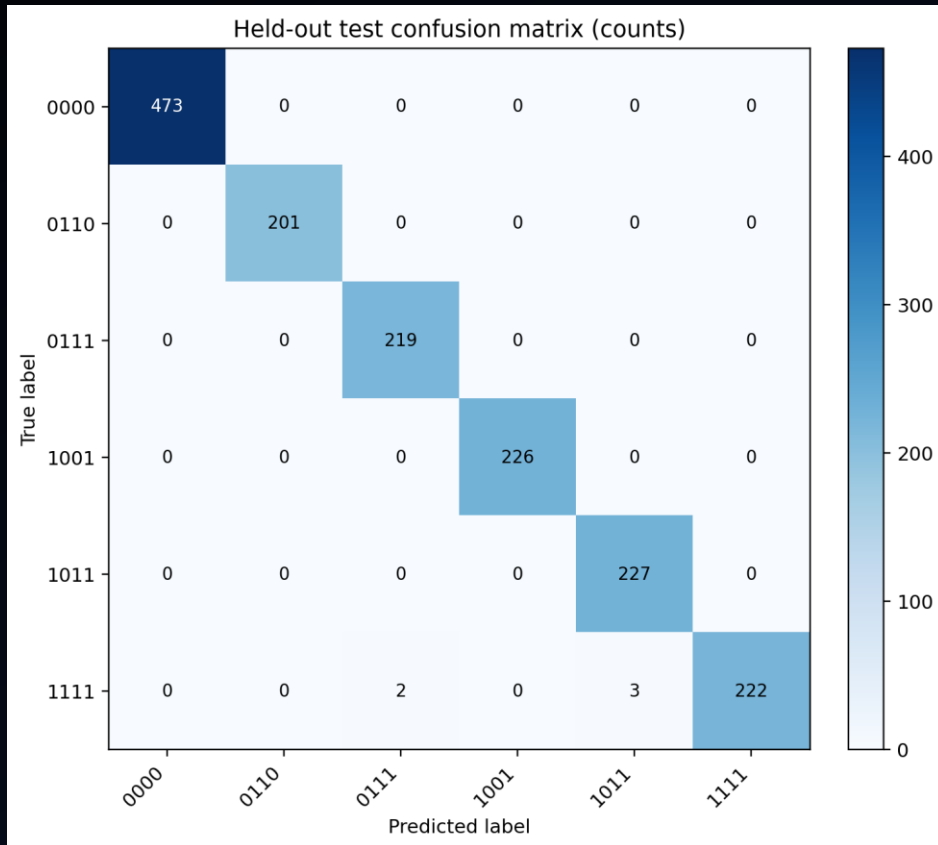
- Relatively small dataset
- Tabular
- Non-linear
- Data appears clean and low-noise
- Simulated circuit data
- Mild class imbalance

- Trial models included top homework performers.
 - kNN
 - RBF SVM
 - Random Forest
 - MLP
- Without proxies, kNN performed best.
 - accuracy=0.8887
 - balanced_accuracy=0.8691,
 - f1=0.8691
- With proxies, **Random Forest** performed excellently, making it the final model choice.

RESULTS AND OBSERVATIONS

- Model training highlights:
 - Tuned with a grid search
 - 300 trees
 - Mean depth: 15
 - Mean leaf count: 112
 - Balanced subsample class weight
 - OOB score: 99.7%
 - Perfect training metrics
 - 99.7% validation accuracy.
- Test performance:
 - Accuracy: 99.68%
 - Balanced accuracy: 99.63%
 - Precision: 99.63%
 - Recall: 99.63%
 - F1: 99.63%
 - Only 5 wrong predictions

Model Performance Visuals



INTERPRETATION

- Feature representation matters significantly.
- Having real-world understanding helps motivate useful decisions.
- Higher complexity does not necessarily mean superior performance
 - Random forest beat MLP.
- Visualizing feature-space geometry helps inform transformation decisions.
- Multiple models can perform well on the same task.
 - Other factors should also influence final decision.
- Validation and test results show that random forest was a good choice for this problem.

CONCLUSION

- Main takeaway: spend more time thinking about the data and problem to make good model choices before spending a lot of effort tuning a model that was a sup-par decision.
- While real-time sensors are actually used to detect faults, applying machine learning techniques still revealed behavioral patterns and performance structure that is useful in understanding fault detection.
- Sometimes decision trees outperform neural networks.